Factors Influencing Topical Glucocorticoid Potency and Side Effects

Factors that influence the potency and side effects of topical glucocorticoids include: - Molecular structure of the compound
- Vehicle (ointment, cream, lotion)
- Amount of medication applied
- Duration of application
- Use of occlusion
- Host factors such as age, body surface area, weight, skin inflammation, anatomic location of treated skin, and individual differences in cutaneous or systemic metabolism

Side effects are directly related to the potency of the glucocorticoid and the duration of use. Clinicians must balance the need for effective treatment with the risk of adverse effects.

Ointments have greater occlusive properties compared to creams, which enhances epidermal penetration and increases systemic absorption.

Side Effects of Topical Glucocorticoids

Side effects can be categorized as **local** or **systemic**.

Local Side Effects

- Striae
- Skin atrophy
- Perioral dermatitis
- Acne rosacea

Systemic Side Effects

- Hypothalamic-pituitary-adrenal (HPA) axis suppression

The risk of adrenal suppression is highest in infants and young children. However, studies have shown that **fluticasone propionate 0.05% cream**, a mid-potency steroid, is safe and effective in children as young as 3 months when used on the face or large body areas for up to 1 month. It is approved for use in children aged 3 months and older for up to 4 weeks. Fluticasone lotion is approved for children 12 months and older. Mometasone cream and ointment are approved for children aged 2 years and older.

Topical Corticosteroids for Maintenance Therapy

Even clinically normal skin in atopic dermatitis (AD) shows immunologic dysregulation. Controlled studies have demonstrated that after achieving control with daily topical corticosteroids, long-term remission can be maintained with **twice-weekly application of fluticasone** to previously affected areas.

Topical Calcineurin Inhibitors

Topical **tacrolimus** and **pimecrolimus** are nonsteroidal immunomodulators.

- **Tacrolimus 0.03% ointment**: approved for intermittent treatment of moderate to severe AD in children ≥2 years
- **Tacrolimus 0.1% ointment**: approved for adults
- **Pimecrolimus 1% cream**: approved for mild to moderate AD in patients ≥2 years

Both agents have demonstrated efficacy and a favorable safety profile: - Tacrolimus ointment: safe for use up to 4 years
- Pimecrolimus cream: safe for use up to 2 years

Side Effects

- Transient burning or stinging sensation at the application site (common)
- No association with **skin atrophy**, making them ideal for sensitive areas such as the face and intertriginous regions

Safety Considerations

- No increased risk of viral superinfections, including eczema herpeticum, has been observed in ongoing surveillance
- Long-term safety beyond several years remains undetermined
- Rare cases of skin malignancy and lymphoma have been reported with tacrolimus, but the scientific consensus considers the evidence low quality and not causally linked

Identification and Elimination of Triggering Factors

General Considerations

Patients with AD are more sensitive to irritants than unaffected individuals. Identifying and eliminating triggers is essential to interrupt the **itch-scratch cycle**.

Common Irritants to Avoid

- Soaps and detergents with high defatting activity
- Contact with chemicals
- Tobacco smoke
- Abrasive clothing (e.g., wool, synthetic fabrics)
- Extreme temperatures and humidity

- Alcohol and astringents in toiletries (can be drying)

Skin Care Recommendations

- Use soaps with **minimal defatting action** and **neutral pH**
- Launder new clothing before wearing to reduce exposure to formaldehyde and other chemicals
- Residual laundry detergent can be irritating; use **liquid detergent** and add a **second rinse cycle** for better removal

Environmental Modifications

- Maintain optimal **temperature and humidity** to minimize sweating and dryness
- Encourage normal physical activity in children, with appropriate skin protection

Every effort should be made to support a normal lifestyle while minimizing exposure to known triggers.